



Barcelona School of Economics

Master's Thesis

Data Science Methodology

Research Question

Can class-conditional Denoising Diffusion Probabilistic Models generate synthetic satellite images that improve classification performance on small datasets like EuroSAT compared to traditional data augmentation methods?

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Abstract

Data augmentation is crucial for improving deep learning model performance, particularly in scenarios where training data is scarce. While traditional geometric transformations have proven effective, generative models, specifically Denoising Diffusion Probabilistic Models (DDPMs), offer new possibilities for synthetic data generation. This thesis explores whether class-conditional DDPM-generated satellite images can enhance classification performance on limited datasets, using EuroSAT as a benchmark. We implemented a class-conditional DDPM with a U-Net architecture to generate synthetic satellite images across 10 land cover classes. Classification performance was evaluated using ResNet-18 on datasets containing 100%, 50%, and 10% of the original 27,000 images, with synthetic augmentation ratios ranging from 5%-50%).

Results indicate that geometric augmentation consistently outperformed synthetic approaches. For the 10% dataset, geometric augmentation achieved significant improvements (+0.79 percentage points, $p < 0.001$), whereas synthetic data augmentation often led to performance degradation, particularly at higher ratios (30%-50% synthetic data showing drops of 0.46-1.53 percentage points). Training exclusively on synthetic data yielded only 83.59% accuracy compared to 98.94% with real data, indicating substantial distributional differences. However, a per-class analysis revealed nuanced results: Hybrid synthetic-geometric approaches yielded optimal results for five classes, notably AnnualCrop (+2.65 percentage points). This suggests that synthetic augmentation effectiveness is highly class-dependent and may benefit from targeted application strategies. Overall, our findings indicate that geometric augmentation is the most effective approach for satellite image classification, however synthetic DDPM-generated images combined with geometric augmentation can lead to class-specific improvements for certain class (land cover) types.

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1 Introduction

In the evolving landscape of machine learning and artificial intelligence, it has become a prominent property that effective models depend strongly on the amount of data available to train it. A comprehensive dataset ensures generalization and the ability of handling complex tasks in any setting.

Deep learning models such as diffusion models have become increasingly deployed in numerous sectors to artificially create images. [1, 2] We will thus analyze how the generation of synthetic images using diffusion can help a predictive model in terms of performance. In scenarios where acquiring large labeled datasets is costly and time-consuming — such as medical imaging [3] or satellite data, generative models offer a viable solution — augmenting the dataset with artificial, but realistic, synthetic data.

For this reason, this thesis explores how the use of synthetic data generated by diffusion models can affect the performance of image classifiers. Our main aim is to investigate how the ratio of synthetic data used affects the performance; We employ three diffusion models, each trained on datasets of varying sizes, based on the intuition that sparser datasets generally benefit more from augmentation. Thus, performance gains are expected to be greater for smaller datasets. We focus on the EuroSAT dataset to classify satellite images, and train diffusion models on 100%, 50% and 10% of the data. We then conduct experiments using no augmentation, geometric augmentation, different ratios of synthetic data augmentation, and evaluate how each of these settings impact the performance of a classifier (ResNet-18) on the test accuracy.

This thesis is structured as follows:

In Section 2, we review existing work on diffusion models and particularly regarding generative augmentation. In section 3 we provide an overview of the theoretical background on diffusion models. Section 4 outlines our experimental setup, including dataset choices, classifier design, diffusion model training, and evaluation metrics. Section 5 presents and analyzes the experimental results. Section 6 discusses potential reasons and limitation regarding these findings. Finally, Section 6 concludes with key findings and a discussion of future research directions.

2 Related Work

Data augmentation using generative models, has become a significant area of research for enhancing machine learning model performance, especially in scenarios with limited data.

Several generative models have been developed and explored in detail, some with the goal of enlarging limited datasets and improving model performance in classification or segmentation tasks. Among the most widely studied approaches are Variational Autoencoders (VAEs) [4], Generative Adversarial Networks (GANs) [5], and, more recently, diffusion probabilistic models, which were first introduced by Sohl-Dickstein et. al (2015) [6], and improved by Ho. et al (2020) [7]. Among these, diffusion models, particularly the Denoising Diffusion Probabilistic Models (DDPMs) introduced by Ho et al. (2020), form the core focus of this thesis due to their capabilities in high-fidelity image

generation and their suitability for investigating fundamental augmentation principles using a simple pipeline.

This section presents an overview of the most relevant generative augmentation techniques, with a particular emphasis on the role of diffusion models. Our thesis does not focus on state-of-the-art (SOTA) diffusion models, namely Stable Diffusion, but instead we focus on the original diffusion model implementation by Ho et al.[7]. This is because implementing a Stable Diffusion pipeline involves significant effort and careful planning of computational resources, placing it outside the scope and cost-limit of this thesis. Nonetheless, the following review presents the relevant related work on diffusion models focused on data augmentation.

(a) Synthetic Image Creation

Image augmentation is a common pre-processing step in machine learning workflows, as it allows for enhancing and enlarging data without adding additional independent training data. The goal is to enhance the ability of the model to learning about the training data with enhanced variety in the training data. This has been shown to work well in many research papers.

The first major breakthrough in generating synthetic images during the rise of deep learning was the development of Generative Adversarial Networks (GAN). These models are based on convolutional neural networks, more specifically on two neural networks that consist of a generator and a discriminator. The generator tries to produce realistic images, and the discriminator learns to distinguish between real and synthetic ones [8].

GANs have demonstrated wide applicability across various domains, particularly where data scarcity still poses an important challenge. Adedji et al. (2022) applied GANs to the EuroSAT dataset, evaluating their effectiveness in enhancing the generalization performance of classification models [9].

Their results show that the inclusion of GAN-generated satellite images during training led to modest but consistent improvements in classification accuracy. This has been found to be independent of the architecture used but nevertheless highlights the potential of generative models for domain-specific augmentation such as satellite images.

However, despite these successes, GANs are known for training instability and mode collapse [10]. So, while they can deliver promising results, other augmentation techniques such as diffusion models can provide even better results. Dhariwal and Nichol [11] showed that diffusion models (specifically DDPMs) not only produce higher-quality images across multiple resolutions on the ImageNet benchmark, but also provide better coverage of the data distribution. They further introduce classifier guidance, which enables controlled, class-conditional generation without compromising image realism. These improvements motivate the use of diffusion-based models for augmentation in this thesis.

(b) Improving Model Performance using Diffusion Models

In recent years, a growing amount of research has focused on diffusion models and their capabilities in generating high-quality synthetic data. Capel & Dumas demonstrate

how diffusion models can be effectively leveraged to structure latent spaces and generate artificial images. [12] in the context of energy market forecasting.

Building on this, researchers have also explored the idea of using diffusion to enhance model performance in various other domains.

For instance, diffusion models have been used for time series forecasting in power systems, showing improvements in energy demand prediction accuracy [13]. In the medical domain, Mohamed Akrouf et al [14] apply diffusion-based data augmentation for skin disease classification and demonstrate consistent model improvement when artificially increasing the dataset size. Similarly, Yanghao Wang and Long Chen use a diffusion framework specifically designed to boost classification in data-scarce scenarios.[15] Across these studies, diffusion-based augmentation consistently proves to be a promising strategy, offering improvements in model performance. Clearly, research in the area of medicine is not scarce anymore. Given the wealth of existing studies and the specialized domain knowledge required for applying diffusion models to medical datasets, we opted to base our experiments on the EuroSAT dataset. This choice aligns with the approach taken by Sousa et al. in their paper "Data Augmentation in Earth Observation: A Diffusion Model Approach," where they explore the application of diffusion models for augmenting satellite imagery.

In their work, Sousa et al. propose a four-stage augmentation pipeline that utilizes meta-prompt generation, vision-language models for captioning, fine-tuning of diffusion models, and iterative data augmentation. Their method demonstrates improvements in classification accuracy, particularly by enhancing the semantic diversity of the augmented data.[16]

However, their study does not delve deeply into how varying the amount of synthetic data impacts classification performance, and neither does it compare different generative model configurations. To address these gaps, our research focuses on systematically analyzing the effects of different ratios of synthetic to real data and comparing the efficacy of class-conditional diffusion models versus multiple class-specific diffusion models. Through this investigation, we aim to complement existing research by providing insights into how the quantity and configuration of synthetic data influence model performance.

3 Theoretical Background

Since the aim of this thesis is to analyze how data augmentation using diffusion models affects classification performance, we begin by outlining the theoretical foundations of diffusion models. This section provides an overview of their core principles, training mechanisms, and practical applications.

In general, diffusion models are a class of generative models that operate by learning to reverse a gradual noising process applied to input data. At the very high level, they consist of two processes:

- (1) **The forward process:** In this process, also known as the diffusion process, Gaussian noise is added incrementally to purposely distort the input.
- (2) **The reverse process:** A neural network is applied to learn how to denoise the distorted input back to its original data distribution.

(a) Diffusion-based generative modeling

There are several types of diffusion models, and the following introduces the key theoretical components of the models.

The first widely adopted formulation of diffusion models is the Denoising Diffusion Probabilistic Model (DDPM). As briefly outlined above, these models operate by learning to reverse a gradual noising process applied to the data. What sets DDPMs apart is their training objective: minimizing the variational lower bound on the data (log) likelihood. This gives them a solid theoretical foundation, which we will explore more deeply in the following section. Despite their computational cost due to the iterative denoising process, DDPMs have achieved state-of-the-art image generation quality and serve as the building block of many modern diffusion-based architectures. Their release marked an important moment in the evolution of practical generative modeling. It is especially suited and embraced by the creative industries but their flexible nature has also enabled adaptation to diverse domains, including applications in medicine and beyond.

A closely related formulation to DDPMs is that of Score-Based Generative Models (SGMs). It focuses not on reversing the noise process, but instead on learning the score function — the gradient of the log-probability density of the data. This involves solving stochastic differential equations that incorporate these learned gradients. Training is typically performed using denoising score matching, and image generation relies on Langevin dynamics or similar sampling techniques.[17]

This can be further generalized through the use of Stochastic Differential Equations (SDEs). This is basically continuous-time formulation of both the forward diffusion and reverse generation processes. The approach unifies DDPMs and SGMs under the umbrella of stochastic processes and allows for flexible sampling procedures by understanding the objective in general terms. It thus strengthens the theoretical foundation and opens new way of improving sampling efficiency and model expressivity. [18].

Nonetheless, DDPMs are most commonly used as they offer a well-understood and

stable training framework and are widely supported in existing libraries such as HuggingFace’s `diffusers`.

Due to its efficiency, flexibility, and open-source nature, we use DDPM as the foundation for our generative augmentation method in this thesis.

In the following subsections, we detail the mathematical structure of diffusion models and their training objectives. It is inspired by the structured presentation of concepts in Prince’s work on deep learning [19] and Luo’s unified perspective on diffusion models [20]. We adopt a similar approach to walk through key theoretical formulations that support our experimental design.

(b) Set-up and Generative Modeling

Before diving into the mathematical formulation of diffusion models, we begin with a foundational understanding of the input and output structure.

Our inputs are images, so it is essential to understand how these are represented. A color image consists of three channels: red, green, and blue - also known as RGB. Each channel corresponds to a matrix of pixel values, where the intensity of each pixel is encoded as a value between 0 and 255 (or normalized between 0 and 1). For an image of size $H \times W$, the full representation is an object of shape $3 \times H \times W$.

(i) Latent Variable Models

Clearly, images often lie in a very high-dimensional spaces. For instance, a 256×256 RGB image lies in $\mathbb{R}^{196,608}$. However, practically images do not occupy this entire space; they lie on a much lower-dimensional manifold. Sampling a random vector from $\mathbb{R}^{196,608}$ would almost never yield a meaningful image.

Thus, we assume that the observed data $\mathbf{x}$ is generated from a lower-dimensional latent variable $\mathbf{z}$. Under this condition, we want to use this space (representing the data) and train a neural network in such a way that we can transform a sample in the latent space to the original space. The idea of latent space can be intuitively explained by utilizing Plato’s Allegory of the Cave: A person is captured with chains inside a cave and can only see the shadows projected onto the wall by objects behind them. The shadows are low-dimensional representations while the real objects of higher dimensions causing the shadows remain hidden.

Overall, this is what we call *latent variable models*, where we model the data as observed as joint distribution $p(\mathbf{x}, \mathbf{z})$. Building on this, we aim to recover the marginal likelihood $p(\mathbf{x}) = \int p(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$.

(ii) Generative Modeling Objective

As we have seen, we are given observed samples $\mathbf{x}$ from some true data distribution $p(\mathbf{x})$, and we want to approximate this as best as possible - we want to maximize the likelihood of observing the data. In the case of diffusion models, when we aim to learn the data distribution $p(\mathbf{x})$, we specifically refer to the distribution of clean data at timestep $t = 0$, denoted $p_\theta(\mathbf{x}_0)$. We can view the denoising process as a latent variable

model, where the clean image x_0 is generated from a series of latent noisy variables $x_1, \dots, x_T$. Thus, the quantity we aim to model is the marginal likelihood $p_\theta(\mathbf{x}_0)$, the likelihood of reconstructing the clean image from noise.

In general, in latent variable models, this involves learning a mapping from latent variables $\mathbf{z}$ to observations $\mathbf{x}$. Once we have learned the latent representation, we can generate new samples from our approximation, completing our generative modeling objective.

(iii) Evidence Lower Bound (ELBO)

However, since maximizing the marginal likelihood $p(\mathbf{x})$ or put differently, $p_\theta(\mathbf{x}_0)$ directly is typically intractable to compute directly due to the integral over $\mathbf{z}$, we optimize a lower bound called the Evidence Lower Bound (ELBO).

Mathematically, we can write $p(\mathbf{x})$ as $p(\mathbf{x}) = \int p(\mathbf{x}, \mathbf{z}) d\mathbf{z}$ to see how maximizing $p(\mathbf{x})$ leads to the ELBO:

$$\log p(\mathbf{x}) = \log \int p(\mathbf{x}, \mathbf{z}) d\mathbf{z} \quad (1)$$

$$= \log \int \frac{q_\phi(\mathbf{z}|\mathbf{x})}{q_\phi(\mathbf{z}|\mathbf{x})} p(\mathbf{x}, \mathbf{z}) d\mathbf{z} \quad (2)$$

$$= \log \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \left[\frac{p(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] \quad (3)$$

$$\geq \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \left[\log \frac{p(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] = \mathcal{L}_{\text{ELBO}}(\theta, \phi) \quad (4)$$

where the last step in the inequality results from applying Jensen's inequality to the log of an expectation. However, this derivation does not clearly explain why we want to maximize the ELBO to maximize the objective. To do this, we can rewrite $p(\mathbf{x})$ using the chain rule of probability.

$$p(x) = \frac{p(x, z)}{p(z|x)} \quad (5)$$

To derive the Evidence Lower Bound (ELBO), we begin by expressing the log-likelihood of the data $\mathbf{x}$ using the marginalization over the latent variable $\mathbf{z}$:

$$\log p(\mathbf{x}) = \log p(\mathbf{x}) \int q_\phi(\mathbf{z}|\mathbf{x}) d\mathbf{z} \quad (6)$$

$$= \int q_\phi(\mathbf{z}|\mathbf{x}) \log p(\mathbf{x}) d\mathbf{z} \quad (7)$$

$$= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [\log p(\mathbf{x})] \quad (8)$$

Using equation 5, we substitute this into the log:

$$= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \left[\log \frac{p(\mathbf{x}, \mathbf{z})}{p(\mathbf{z}|\mathbf{x})} \right] \quad (9)$$

We now multiply and divide the argument of the log by $q_\phi(\mathbf{z}|\mathbf{x})$:

$$= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \left[\log \frac{p_\theta(\mathbf{x}, \mathbf{z}) q_\phi(\mathbf{z}|\mathbf{x})}{p(\mathbf{z}|\mathbf{x}) q_\phi(\mathbf{z}|\mathbf{x})} \right] \quad (10)$$

Then split the log into two terms, recognizing that the second term is the KL divergence:

$$= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \left[\log \frac{p(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] + \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \left[\log \frac{q_\phi(\mathbf{z}|\mathbf{x})}{p(\mathbf{z}|\mathbf{x})} \right] \quad (11)$$

$$= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \left[\log \frac{p(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] + D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}|\mathbf{x})) \quad (12)$$

Finally, since KL divergence is non-negative, we get to exactly the lower bound, exactly as before. In fact, we can observe how the KL divergence is the term getting absolved when applying the Jensen Inequality as above.

$$\log p(\mathbf{x}) \geq \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \left[\log \frac{p(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] = \mathcal{L}_{\text{ELBO}} \quad (13)$$

Now notice that we have

$$D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}|\mathbf{x})) = \log p(\mathbf{x}) - \mathcal{L}_{\text{ELBO}} \quad (14)$$

where $\log p(\mathbf{x})$ is a constant with respect to ϕ . Thus, maximizing the ELBO is equivalent to minimizing the divergence between the approximate and true posteriors. We can see how we can use the ELBO as objective to approximate the intractable posterior.

(c) The diffusion Process

Before seeing how exactly this is applied in latent diffusion models, we need to understand to what this maximization is actually applied. Therefore, we need to understand how the forward and reverse process shortly outlined above work in detail.

(i) The forward Process

The forward process, also called the diffusion process, is where we gradually add noise to our input images. Over a series of T timesteps, noise is added in a controlled manner such that the original image becomes progressively more corrupted, until it turns into pure Gaussian noise.

This process can be described mathematically in a parameterized form as follows:

$$\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (15)$$

Here, β_t is a predefined *variance schedule* that controls how much noise is added at each timestep t . Note that the first term, $\sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1}$, ensures that the information in the previous/original image is decreased. The second term, $\sqrt{\beta_t} \cdot \boldsymbol{\epsilon}$, adds Gaussian noise scaled according to β_t .

This way, the variance of the overall image remains approximately constant while the amount of information - the signal gradually disappears. Typically, the number of steps T is set to 1000, which provides a fine-grained noise schedule and improves the quality of reverse sampling (i.e., denoising).

This process can equivalently be written in terms of a one-step conditional distribution:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1}, \beta_t \cdot \mathbf{I}) \quad (16)$$

That is, each noisy image $\mathbf{x}_t$ is sampled from a Gaussian distribution with a mean that is a scaled version of the previous step $\mathbf{x}_{t-1}$, and variance β_t . [19]

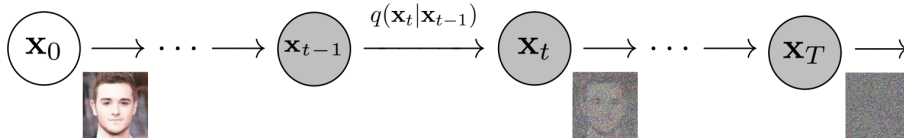


Figure 1: Illustration of the forward diffusion process. At each timestep, Gaussian noise is added to the image, gradually corrupting it until it resembles pure noise at timestep T . Adapted from Aristimuño (2023) [21].

Overall, we can see that the process is Markovian, because each state $\mathbf{x}_t$ depends only on the immediately previous state $\mathbf{x}_{t-1}$ and not on any earlier ones - the forward process defines a Markov chain. Thus, the joint distribution over all latent variables $\mathbf{x}_{1:T}$ given the original input $\mathbf{x}_0$ can be written as:

$$q(\mathbf{x}_{1:T} | \mathbf{x}_0) = q(\mathbf{x}_1 | \mathbf{x}_0) \prod_{t=2}^T q(\mathbf{x}_t | \mathbf{x}_{t-1}) \quad (17)$$

[19, 20]

Note on notation: In this thesis, we follow the convention of denoting the noised data at timestep t as $\mathbf{x}_t$. In some literature — particularly when dealing with latent spaces (i.e Latent Diffusion Models), this variable may instead be denoted $\mathbf{z}_t$. While the principles remain the same, we use $\mathbf{x}_t$ to keep the notation consistent and emphasize that we talk about image-like data.

(ii) The Reverse Process

The reverse process $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$, also known as the denoising process, is the generative process that attempts to reverse the forward diffusion by gradually removing noise from the fully corrupted image $\mathbf{x}_T$ step by step, until we reach a clean sample $\mathbf{x}_0$. Unlike the forward process, which is fixed and analytically defined, the reverse process is learned and modeled using a neural network.

The goal of this process is to approximate the true reverse conditional distribution $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$, which is typically intractable. To compute this, we would need to integrate over all possible clean images $\mathbf{x}_0$ that could have been produced $\mathbf{x}_t$. Mathematically, this involves computing:

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t) = \int q(\mathbf{x}_{t-1} | \mathbf{x}_0, \mathbf{x}_t) q(\mathbf{x}_0 | \mathbf{x}_t) d\mathbf{x}_0 \quad (18)$$

To address this, we introduce a neural network that learns to predict the noise added at each timestep. Since each step in the forward diffusion process involves adding Gaussian noise, it follows that the true reverse conditional distribution is also Gaussian.

Therefore, we define the learned reverse process as a parameterized Gaussian:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t)) \quad (19)$$

Where $\boldsymbol{\mu}_\theta$ and $\boldsymbol{\Sigma}_\theta$ are the predicted mean and variance at each timestep, learned by the model. [19, 2],

The neural network used in the reverse process (typically a U-Net) is trained to estimate the noise component ϵ present in the noised image $\mathbf{x}_t$ at each timestep t . This is done by providing the model with the noisy image $\mathbf{x}_t$ and the timestep t as input, and minimizing the difference between the predicted noise $\epsilon_\theta(\mathbf{x}_t, t)$ and the true noise that was added during the forward process.

This objective can be theoretically derived as a special case of maximizing the Evidence Lower Bound (ELBO), as previously introduced for general latent variable models. In the case of diffusion models, the ELBO takes on a specific structure, which we present in the next section.

Overall, the model starts from a random noise sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$ and applies the learned reverse transitions $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ recursively, denoising step by step until it reaches a clean image sample $\mathbf{x}_0$.

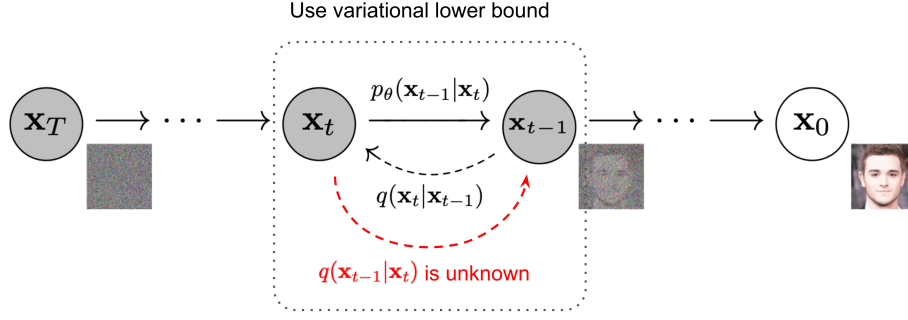


Figure 2: Illustration of the reverse diffusion process. The model is trained to predict the clean images. Adapted from Aristimuño (2023) [21].

(d) ELBO in Diffusion

As shown in the unified perspective on diffusion models by Luo [20], the ELBO in the case of a diffusion model, can be written as a sum of three interpretable terms:

$$\begin{aligned} \mathcal{L}_{\text{ELBO}} = & \underbrace{\mathbb{E}_{q(x_1|x_0)}[\log p_\theta(x_0 | x_1)]}_{\text{reconstruction term}} - \underbrace{D_{\text{KL}}(q(x_T | x_0) \parallel p(x_T))}_{\text{prior matching term}} \\ & - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(x_t|x_0)}[D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \parallel p_\theta(x_{t-1} | x_t))]}_{\text{denoising matching term}}. \end{aligned} \quad (20)$$

In Eq. (20), the first term—the reconstruction term—is the expected log-likelihood of denoising a single step. In practice, it is estimated efficiently via a Monte Carlo approximation.

The second term, the prior-matching term, measures how closely the noisy latent at the final timestep resembles a the standard Gaussian (prior); minimizing it encourages the forward noising process to produce samples that the reverse process can reliably map back to clean data.

Finally, the summation of KL divergences over $t = 2 \dots T$ can be called the denoising-matching term: each individual KL quantifies how well the model’s predicted reverse distribution $p_\theta(x_{t-1} | x_t)$ aligns with the true “ground-truth” noising distribution, denoted $q(x_{t-1} | x_t, x_0)$ at that timestep, and minimizing it forces the network to match the exact noise signal added in the forward process. Minimizing this is as simple as minimizing the difference between the means of two gaussian distributions q and p . It can be rewritten as in Eq. (21).

(e) Diffusion Objective Revisited

As shown in Eq. (20), diffusion models are trained by minimizing the variational lower bound on the data log-likelihood, using a denoising process that operates step-by-step from noise to data. This formulation effectively trains a model to approximate the true data distribution $p(x_0)$ through a Markov chain of denoising steps:

$$p_\theta(x_0) \approx \int p_\theta(x_0 | x_1) p_\theta(x_1 | x_2) \cdots p_\theta(x_{T-1} | x_T) p(x_T) dx_{1:T}$$

Each term $p_\theta(x_{t-1} | x_t)$ in this chain is modeled by a neural network—typically a U-Net—as discussed in the previous section. The model is trained to reverse the forward noising process $q(x_t | x_0)$ by predicting either the clean image x_0 , the noise ϵ , or the mean of the posterior distribution $\mu_q(x_t, x_0)$.

This gives us a learned generative model that can produce samples $x_0 \sim p_\theta(x_0)$ by sampling from the prior $p(x_T) = \mathcal{N}(0, I)$ and applying the learned reverse denoising steps.

Put differently, the full joint distribution over all variables $x_{0:T}$ of this model can be described as:

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^T p_\theta(x_{t-1} | x_t)$$

Furthermore, each KL term in the sum of Eq. (20) can be minimized by matching the model’s predicted mean to the true noising distribution. [20, 19] In practice one shows that

$$\arg \min_{\theta} D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \parallel p_\theta(x_{t-1} | x_t)) = \arg \min_{\theta} \frac{1}{2\sigma_q^2(t)} \|\mu_\theta(x_t, t) - \mu_q(x_t, x_0)\|_2^2. \quad (21)$$

where

$$\begin{aligned} \mu_q(x_t, x_0) &= \frac{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} x_0, \\ \mu_\theta(x_t, t) &= \frac{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \hat{x}_\theta(x_t, t), \end{aligned}$$

and $\hat{x}_\theta(x_t, t)$ is the network’s estimate of the clean image (or equivalently of the added noise).

With this reformulation we are able to see how the denoising-matching term of the ELBO can be optimized by minimizing the distance between the predicted and target means of two Gaussian distributions. This result is particularly elegant: the initial complex variational inference objective involving distributions over high-dimensional images reduces to the simple and practical mean squared error loss.

(f) Image Prediction

After all, this goes to show that we can use a neural network to predict a clean image given a noisy image and a timestep as input.

In practice, a deep convolutional neural network is used. Because the model must operate at various noise levels and preserve spatial details while being conditioned

on the timestep, the architecture must be both resolution-aware and time-aware. A natural choice that fulfills these requirements is the **U-Net** architecture—originally introduced for biomedical image segmentation, but now widely adopted in diffusion models.

The general structure of such a U-Net is shown in Figure 3. It consists of two main paths: an *encoder* (or contracting path) and a *decoder* (or expansive path).

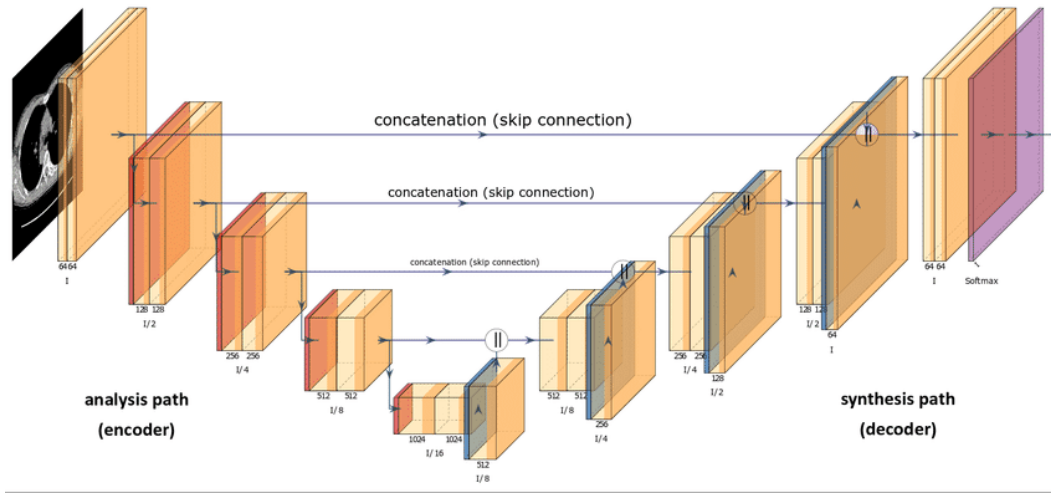


Figure 3: Illustration of a U-Net architecture as used in diffusion models. Adapted from [21].

In the **encoding path**, the model applies convolutional blocks consisting of 2 convolutional layers, each followed by non-linear activations (typically ReLU). After that, downsampling layers such as max pooling are included. This reduces the spatial dimensions while increasing the number of feature channels. For instance, an input image of size $256 \times 256 \times 3$ might be downsampled progressively through multiple layers to a latent representation of size $16 \times 16 \times 512$. The final bottleneck layer captures high-level semantic features of the input image.

However, due to the destructive nature of repeated downsampling, much of the spatial information is lost. To mitigate this, *skip connections* are introduced. These connections pass feature maps from the encoder to the corresponding decoder layer at the same resolution, allowing the network to retain fine-grained details crucial for accurate reconstruction.

The **decoding path** performs the inverse operations: upsampling the latent representation back to the original input resolution. Each upsampling step (often implemented using transposed convolutions) is concatenated with the corresponding encoder feature map via the skip connection, followed by convolutional layers and activations. This design enables the network to produce high-fidelity denoised outputs.

As described above, its outputs directly influence the loss. Specifically, the network is trained to **predict either the clean image x_0 , the noise ϵ , or the posterior mean μ_q** , depending on the chosen parameterization. These predictions enter the

ELBO objective as part of the denoising-matching term:

$$\mathbb{E}_{q(x_t|x_0)} [D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \parallel p_\theta(x_{t-1} | x_t))],$$

which is minimized during training. This loss encourages the model to align its reverse-time predictions with the true noising process — effectively learning to undo the noise added at each timestep.

Overall training process for denoising diffusion models can be described by the following steps:

1. Start with a real image $x_0 \sim q(x_0)$ from the dataset.
2. Sample a random timestep $t \sim \mathcal{U}(\{1, \dots, T\})$.
3. Add Gaussian noise to obtain the noisy image x_t using a recursive noise schedule:

$$x_t = \sqrt{1 - \beta_t} \cdot x_{t-1} + \sqrt{\beta_t} \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I})$$

This can alternatively be expressed more practically in closed form (without needing intermediate steps):

$$x_t = \sqrt{\bar{\alpha}_t} \cdot x_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon \quad \text{with } \bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$$

4. Train the neural network $\epsilon_\theta(x_t, t)$ to predict the added noise ϵ .
5. Minimize the mean squared error loss:

$$\mathcal{L}_{\text{simple}}(\theta) = \mathbb{E}_{x_0, \epsilon, t} [\|\epsilon_\theta(x_t, t) - \epsilon\|^2]$$

[7]

(g) Conditional Diffusion Models

While standard diffusion models learn to generate data purely from noise, in many tasks—such as in our experiments—we want to control what is generated. For example, we may wish to generate an image belonging to a specific class or to modify an input image in a guided manner. To achieve this, conditional diffusion models extend the basic framework by incorporating additional information—a conditioning variable—into the denoising process.

We can recall from Eq. (20) that the training objective approximates the log-likelihood of the data under the learned model $p_\theta(x_0)$, by decomposing it into denoising subproblems via the ELBO. In other words, we aim to model:

$$p_\theta(x_0) = \int p_\theta(x_0 | x_T) p(x_T) dx_T$$

However, in the conditional case, we wish to sample from a conditional distribution:

$$p_{\theta}(x_0 | y) = \int p_{\theta}(x_0 | x_T, y) p(x_T) dx_T$$

This modifies the reverse diffusion process such that each denoising step becomes conditioned on y :

$$p_{\theta}(x_{t-1} | x_t, y)$$

There are multiple strategies to implement this conditioning in practice[19]:

- Classifier Guidance: uses a separate pre-trained classifier to guide the denoising process toward samples matching a desired label.
- Classifier-Free Guidance avoids the need for an external classifier by training the model to denoise both with and without conditioning. During inference, the model combines both outputs using a *guidance scale*.

However, in our experiments, we follow a more simple but effective strategy: we directly embed the class label and provide it as additional input to the denoising model (the UNet). Specifically, we extend the UNet architecture with a learned class embedding layer, and pass the embedded label along with the noisy input image x_t and the timestep t during training. More detail on this can be seen in the experiments section, where we described the diffusion model applied in more detail.

This way, the model learns to denoise not only based on the noisy image and the timestep, but also conditioned on the desired class label. As a result, the trained diffusion model can generate samples corresponding to specific target classes during inference.

(h) Image Classification

Image classification is a fundamental supervised learning task in which a model learns to assign a category label to an input image. Given a dataset of labeled examples $(\mathbf{x}, y)$, where $\mathbf{x}$ is an image and y is the class label, the goal is to learn a function that predicts the correct label for new, unseen images.

As we can see, this involvement of labeled data matches to conditional generative frameworks as described above. In our case, conditional diffusion allows us to model $p(x_0 | y)$. We can then apply targeted image generation conditioned on the desired class label.

The model is trained to minimize a loss function (i.e the categorical cross-entropy) between the true labels and the predicted class probabilities. The effectiveness of this depends strongly on the amount and quality of available training data. In low-data regimes, the model may fail to generalize well, leading to overfitting or poor decision boundaries.

Now by training a classifier on varying combinations of real and synthetic data, we aim to answer our core research question: *Does diffusion-generated data improve classification performance compared to using only real data?*

(i) Model Selection for Earth Observation Classification

In computer vision, numerous deep learning architectures have been developed for image classification tasks, each with distinct characteristics and performance profiles. For Earth Observation applications like land cover classification, the selection of an appropriate architecture is crucial given the unique challenges of satellite imagery.

Convolutional Neural Networks (CNNs) remain the foundation of modern image classification approaches. The earliest successful architecture, AlexNet, demonstrated the power of deep learning for image recognition, while VGG networks introduced the concept of using smaller filters with greater depth. However, these early architectures suffer from the vanishing gradient problem when scaled to greater depths, limiting their effectiveness for complex classification tasks.

ResNet (Residual Network) architectures addressed this fundamental limitation through the introduction of skip connections, allowing signals to bypass layers during backpropagation. This innovation enabled the training of much deeper networks, with variants ranging from 18 to 152 layers. For satellite imagery classification specifically, the depth of these networks provides an advantage in capturing multi-scale features necessary for distinguishing between similar land cover types.

Following the approach of Adedeji et al. (2022), we initially considered Wide ResNet50 as our classification architecture, as it demonstrated superior performance in their study. However, due to computational limitations, we ultimately implemented ResNet18 for our experiments. While ResNet18 is shallower (18 layers compared to 50) and lacks the increased width of Wide ResNet models, it offers several advantages for our specific use case. ResNet18 maintains the crucial residual learning framework that enables effective gradient flow during training, while requiring significantly fewer parameters (11.7 million versus 68.9 million in ResNet50). This lighter architecture allows for faster training iterations and reduced computational requirements, which is particularly important when evaluating multiple augmentation strategies. Despite its more compact structure, ResNet18 has demonstrated strong performance on image classification tasks, especially when leveraging transfer learning from ImageNet pretraining. More specifically, ResNet18 turned out to be more than twice as fast while barely degrading accuracy. This architecture provides an appropriate balance between computational efficiency and model capacity for our comparative study of diffusion-based data augmentation techniques.

4 Experiments

(a) Overview

This section outlines the methodology and experimental procedures used to evaluate the impact of diffusion model-based data augmentation on satellite image classification. The primary objective of this study is to determine whether incorporating synthetic images generated using a DDPM can enhance the performance of a downstream classification task. While prior research has demonstrated this, most approaches rely on more advanced, state-of-the-art text-to-image diffusion models, often requiring elaborate prompt engineering to guide image synthesis. In contrast, our study simply aims to exploit the simplicity of DDPM and investigates whether the ratio of synthetic data added influences the performance of the classifier.

To achieve this, we first introduce the EuroSAT dataset. We then describe the implementation and training specifics of our chosen diffusion model and outline the specifics of the Wide ResNet18 classifier that serves as our evaluation tool for the experiments that we run. Finally, we lay out the experimental design, which focuses on comparing the classifier’s performance under different training data compositions. This includes the baseline of only real data, standard data augmentations (e.g. geometric transformations), and varying proportions of synthetic data augmentations.

(b) Dataset

For this project, we utilize the EuroSAT dataset [22]. The dataset consists of 27,000 labeled and geo-referenced images derived from Sentinel-2 satellite imagery. Sentinel-2, operated by the European Space Agency (ESA) since its launch in 2015, provide high-resolution multispectral images used for various earth observation tasks.

The EuroSAT dataset is available in two forms:

- An RGB version consisting of standard three-channel color images.
- A multispectral version that includes additional spectral bands such as near-infrared and shortwave-infrared.

Although the multispectral version is more information-rich and has been shown in previous works to lead to stable learning, it is significantly slower to train. For this reason, we use the RGB version. As reported in [22], a benchmarking accuracy of up to 98.57% has been achieved using RGB images, and more recently 99.17% in [23].

The RGB images are provided in JPEG format and grouped by classes. There are 10 different land use classes. The classes are slightly imbalanced as seen in the table below. We report the class distributions across the 100%, 50% and 10% datasets.

Table 1: Distribution of classes within the EuroSAT dataset comparing 100% (27,000 images), 50% (13,000 images) and 10% (2,700 images) across 10 distinct land cover categories.

Class ID	Class Name	100% (%)	50% (%)	10% (%)
0	Annual Crop	11.11	11.12	10.78
1	Forest	11.11	11.26	10.04
2	Herbaceous Vegetation	10.68	10.68	11.93
3	Highway	9.26	9.36	10.04
4	Industrial	9.26	9.31	9.30
5	Pasture	7.41	7.38	6.96
6	Permanent Crop	9.26	9.41	9.41
7	Residential	11.11	11.13	11.26
8	River	9.26	9.16	9.41
9	Sea or Lake	11.11	11.19	10.89

Each image has a spatial resolution of 64×64 pixels. A sample of this can be seen in Figure 4.

For pre-processing, the images were split into training and test sets. The label for each image was extracted directly from the filename, as the filenames include the class to which each image belongs.

Additionally, for experiments involving synthetic data, newly generated diffusion images were added to the dataset into the corresponding class folders to match the original EuroSAT structure. During model training, both the real and synthetic images were parsed by extracting labels from their filenames.

Table 1 presents the class distribution across the three datasets used. Since the data was split without explicitly enforcing class balance, slight discrepancies in class proportions are present. However, the differences are minimal and were deemed acceptable for the purposes of this study, as they do not materially affect model performance or comparative analysis.

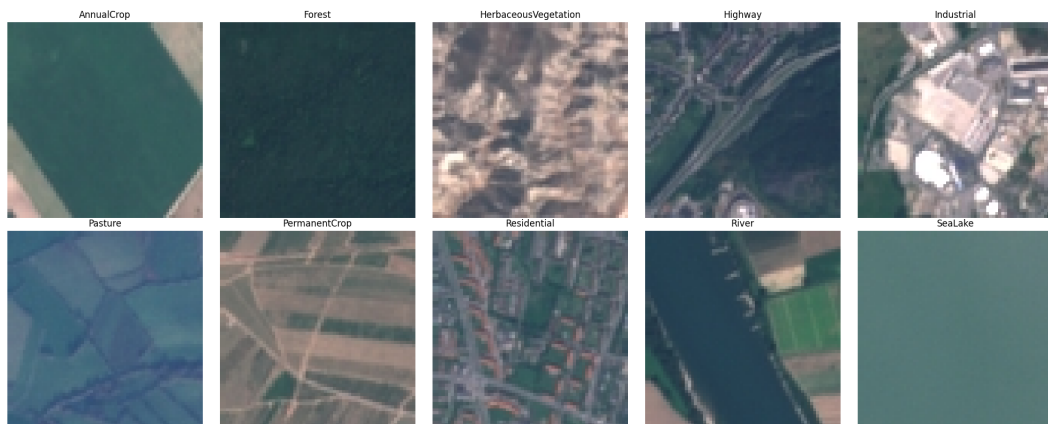


Figure 4: Sample Images from the EuroSat dataset

(c) Implementation of the Diffusion Model

To investigate the impact of diffusion-generated synthetic data on image classification performance, we trained a class-conditional Denoising Diffusion Probabilistic Model (DDPM). A U-Net architecture serves as the noise predictor, trained to reverse a forward diffusion process by removing Gaussian noise in successive denoising steps. We explain the specific architecture and parameters in more detail below.

During training, Gaussian noise is added to each image over 1000 timesteps. The model learns to predict and remove this noise at each step using the Mean Squared Error (MSE) loss, as described in Section (ii). We chose this loss because it provides a simple, and yet stable objective and is commonly used in DDPM-based models.

Conditioning and Embedding

Furthermore, to ensure semantic alignment between generated images and class labels, we employ a class-conditional setup: class labels (mapped to integers from 0 to 9) are embedded as dense vectors and incorporated into the U-Net architecture. These label embeddings are combined with the timestep embeddings, guiding the denoising process at every stage and ensuring that the generated samples are class-consistent.

Timestep embeddings are essential for the model, as the amount of noise in the image varies over time. The model must adapt its denoising behavior depending on the current diffusion step. To enable this, the scalar timestep t is transformed into a high-dimensional vector. These timestep embeddings are then included into the feature maps at multiple layers of the U-Net. This allows the model to recognize the current stage of the denoising process and adjust its predictions accordingly.

This can also be observed in the loss function described above.

We minimize $\mathbb{E}_{x_0, \epsilon, t} [\|\epsilon_\theta(x_t, t) - \epsilon\|^2]$, where x_t is the noised version of the clean image x_0 at timestep t , and ϵ is the noise sampled from a standard normal distribution. However, this t is not fed as a scalar in the model, but is mapped to a high-dimensional vector through an embedding function. This embedding provides the model with temporal context, enabling it to adjust its denoising behavior based on how much noise has been added at that stage of the diffusion process.

Similarly, class embeddings are used to provide the model with information about the desired output class in a class-conditional setup. To keep the architecture simple and efficient, discrete class labels (0-9) are mapped to dense vectors, just like in the case of the time embedding. This conditioning allows the model to generate samples that are consistent with the specified class label, without requiring significant architectural modifications.

In the end, we minimize:

$$\mathcal{L}_{\text{cond}}(\theta) = \mathbb{E}_{x_0, \epsilon, t, y} [\|\epsilon_\theta(x_t, t, y) - \epsilon\|^2],$$

where y is the class label associated with the original image x_0 .

Model Architecture

To provide a comprehensive overview of the architecture used in our model, we summarize the key building blocks of the conditional U-Net in Table 2. The model follows the structure of a typical DDPM where a symmetric encoder-decoder (U-Net) is used to predict the noise added at different diffusion steps.

Each block in the U-Net is composed of multiple so called `ResnetBlock2D` layers. Internally, each ResNet block contains two convolutional layers (`Conv2d`), two group normalization layers (`GroupNorm`), and `SiLU` activations. Additionally, time step information is injected through a `Linear` projection of the timestep embedding, which is added to the output of the first convolution. After Normalization, a drop-out Layer regularizes the model and reduces overfitting: It randomly deactivates a subset of neurons during training and thus encourages the network to be more robust.

In cases where the input and output dimensions differ, a skip connection using a 1×1 convolution (`conv_shortcut`) ensures the correct shape alignment.

In the downsampling path, each `DownBlock2D` consists of two such ResNet blocks, followed by a `Downsample2D` module that reduces spatial resolution using a strided convolution. This combination allows the network to learn increasingly abstract features while shrinking the spatial resolution.

The upsampling path mirrors this structure through `UpBlock2D` modules. Each of these blocks contains three `ResnetBlock2D` layers, where the first layer typically receives concatenated feature maps from the skip connections. At the end of each up block, an `Upsample2D` module is used to increase the spatial resolution via learned convolutions.

At the center of the network lies the bottleneck - another `UNetMidBlock2D`. This block consists of two `ResnetBlock2D` layers and a self-attention layer implemented via multi-head attention. The attention module comprises separate projections for queries, keys, and values, followed by a linear transformation and optional dropout. This allows the model to attend to global spatial relationships that may not be easily captured by local convolutions alone.

This architectural design is consistent with the U-Net description in Section (f), adapted specifically for diffusion-based generation tasks with time and class conditioning. The detailed parameter breakdown and output shapes of each major block are summarized in Table 2.

Table 2: U-Net architecture used in the conditional DDPM model (EuroSAT 64×64)

Block (type)	Layer details	Output shape	# Params
conv_in	Conv2d(3→128)	[B,128,64,64]	3,584
Down Block 1	2×ResNet + Downsample	[B,128,32,32]	870,272
Down Block 2	2×ResNet + Downsample	[B,128,16,16]	2,952,960
Down Block 3	2×ResNet + Downsample	[B,256,8,8]	3,215,104
Down Block 4	2×ResNet + Downsample	[B,256,4,4]	13,383,168
Down Block 5 (Attn)	2×ResNet + Downsample	[B,512,2,2]	9,968,640
Down Block 6	2×ResNet	[B,512,2,2]	4,719,616
Mid Block (Attn)	2×ResNet + Self-Attn	[B,512,2,2]	11,020,288
Up Block 1	3×ResNet + Upsample	[B,512,4,4]	25,968,128
Up Block 2 (Attn)	3×ResNet + Upsample	[B,256,8,8]	27,811,840
Up Block 3	3×ResNet + Upsample	[B,256,16,16]	7,348,480
Up Block 4	3×ResNet + Upsample	[B,128,32,32]	6,364,672
Up Block 5	3×ResNet + Upsample	[B,128,64,64]	1,937,536
Up Block 6	3×ResNet	[B,128,64,64]	1,625,856
conv_out	Conv2d(128→3)	[B,3,64,64]	3,459

The model was implemented using the Hugging Face `diffusers` library and trained for 50 epochs with a batch size of 100-256, using mixed-precision (fp16) training. We adopted a DDPM linear noise scheduler with 1000 diffusion steps. Importantly however, as described in the next section, we only use 250 steps for inference. The optimizer used was AdamW with a learning rate of $1e-4$ and a linear warm-up of 50 steps. Input images were normalized to the $[-1, 1]$ range. The Training loss of the model can be seen in the Figure 5 below.

Importantly, we ran one model with with 10 % and one with 50% of the data for further analysis to re-create the scenario where we only had access to i.e 50% of the data, so not 27 000 images but 13 500 - while keeping all the other parameters (diffusion steps, epochs,..) the same. This could improve the value of the synthetic image creation as described in the section below. As expected, the model trained on the reduced dataset underwent fewer optimization steps per epoch, resulting in a shorter loss curve in terms of training iterations. Nevertheless, both models show signs of convergence, with loss values stabilizing over time. The training loss depicted below is the smoothed average, making absolute comparisons less precise but more easy to see if convergence has been achieved.

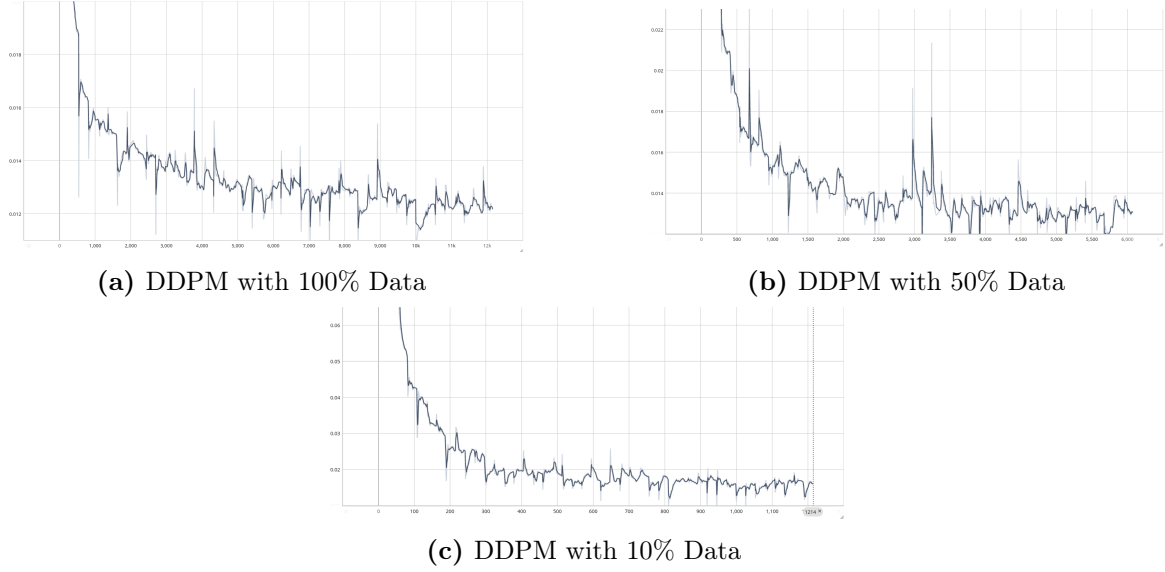


Figure 5: DDPM Loss over Steps/Epochs with Model 1 (a), Model 2 (b) and Model 3 (c)

Inference

At inference time, sampling was performed by iteratively denoising from Gaussian noise while conditioning on a given class label. As described in the experiment section below, we initially generated 128 synthetic samples per class, totaling 1280 images - about 5 percentage of the original training set. We then increase this incrementally.

Qualitatively, the generated samples capture key spatial structures such as agricultural structures, coastlines and urban textures, demonstrating strong semantic alignment. While some classes might seem pretty similar to each other, this seems understandable when taking a closer look at this in the original dataset: For some instances it is difficult to distinguish between labels, for example between *Industrial*, *Residential* (as they both include urban structures that can be quite similar). The visual quality and class alignment of the generated images were verified through manual inspection. A sample set of images created by model 1 and model 3 is shown in Figure 6 and Figure 7, respectively. In practice, the quality of diffusion model generated images is usually evaluated using inception or FID scores, however due to the difficulty in computing these correctly and unbiasedly, we rely on the qualitative assessment.



Figure 6: Sample Images DDPM generated, Model 1

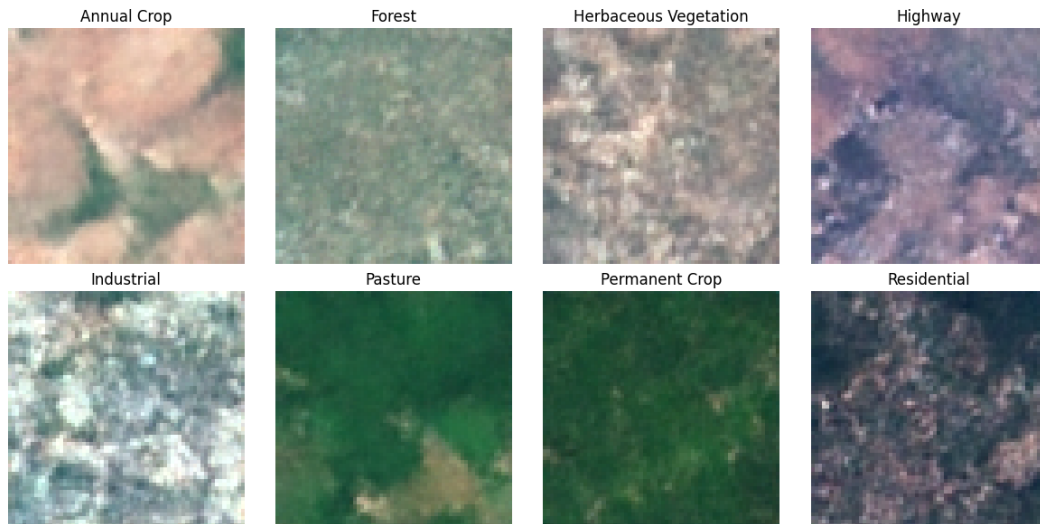


Figure 7: Sample Images DDPM generated, Model 3

To improve efficiency, we reduced the number of inference steps from the default 1000 to 250 after experimenting with it showed minimal degradation in image quality. This led to a significant speed-up in generation time. A qualitative comparison between 250-step and full-step generations is presented in Figure 8.



(a) DDPM model 1 with 1000 steps



(b) DDPM model 1 with 250 steps

Figure 8: Visual quality with 1000 inference steps vs 250 inference steps.

(d) Classification Setup

Model Architecture

As discussed in Section (i) we chose ResNet18 as our classification architecture due to its balance of performance and computational efficiency. Here, we provide more specific details about the architecture implementation: ResNet18 is an 18-layer deep convolutional neural network featuring residual connections that enable effective training of deeper networks.

The standard ResNet-18 architecture for EuroSAT follows this structure:

1. An initial 7×7 convolutional layer with 64 filters and stride 2
2. A 3×3 max pooling layer with stride 2
3. Four residual stages (each containing two residual blocks), with the number of filters doubling and spatial dimensions halving at each stage:
 - **Stage 1:** Two 3×3 conv layers with 64 filters (output: $56 \times 56 \times 64$)
 - **Stage 2:** Two 3×3 conv layers with 128 filters (output: $28 \times 28 \times 128$)
 - **Stage 3:** Two 3×3 conv layers with 256 filters (output: $14 \times 14 \times 256$)
 - **Stage 4:** Two 3×3 conv layers with 512 filters (output: $7 \times 7 \times 512$)
4. A global average pooling layer that reduces spatial dimensions to 1×1
5. A fully connected layer mapping 512 features to our 10 EuroSAT classes

Input Preprocessing:

All images were resized to 224×224 pixels (the standard input size for ImageNet models) and normalized using the ImageNet mean and standard deviation:

$$\text{mean} = [0.485, 0.456, 0.406], \quad \text{std} = [0.229, 0.224, 0.225]$$

Training Protocol:

Our training process utilized the following configuration:

- **Optimizer:** Adam optimizer with a learning rate of $1e-4$
- **Training epochs:** Up to 30 with early stopping (patience of 7 epochs)
- **Learning rate scheduler:** CosineAnnealingLR with $T_{\max} = 20$ and minimum learning rate of $1e-6$
- **Loss function:** Cross-entropy loss for multi-class classification

Evaluation Framework:

We employed a stratified train-test split to ensure balanced class representation, keeping the test transformations consistent across all experiments (resize and normalize only). For each model, we recorded:

- Training and test loss
- Accuracy and weighted F1-score

To ensure reliable comparison, all models were trained with identical hyperparameters, varying only the composition of the training data. The best model for each experiment was selected based on test accuracy, with early stopping applied to prevent overfitting.

(e) Experiment Design

To address our central research question—*Can class-conditional Denoising Diffusion Probabilistic Models generate synthetic satellite images that improve classification performance on small datasets like EuroSAT compared to traditional data augmentation methods?* - we structured our experiments around a consistent classification setup, varying the type and amount of data augmentation used during training. The classifier architecture described in the previous section serves as the foundation for all experiments, ensuring comparability across conditions.

We consider the following scenarios:

- **Full Dataset:** We train the classifier using the full EuroSAT dataset comprising 27,000 RGB images. The original class distribution is slightly imbalanced, as detailed below.
- **Reduced Dataset(s):** We also train the classifier using 10% and 50% of the original dataset (2,700 and 13,500 images), while preserving the original class distribution. This setup allows us to evaluate the effectiveness of synthetic data augmentation in scenarios with limited training data. It corresponds to the scenario of the DDPM model 1, 2, and 3 as described above. The 10% dataset ensures that we account for scenarios where data is scarce (only ≈ 270 images per class).

We consider the following augmentation strategies:

- **Baseline:** The classifier is trained solely on the original (full or reduced) EuroSAT dataset, comprising 27,000, 13,500 or 2,700 RGB images. It serves as the benchmark to measure any improvements potentially achieved by the other methods.
- **Standard Augmentation:** This setup includes classical image augmentation techniques to improve generalization. Specifically:
 - **Random horizontal flips:** 50% chance to mirror the image horizontally, simulating diverse viewing angles.
 - **Random vertical flips:** 50% chance to mirror the image vertically, important for satellite imagery with varying orientations.
 - **Random rotations ($\pm 10^\circ$):** Introduces small rotations to enhance robustness to different spatial alignments.

These transformations preserve the semantic content of the images while encouraging the model to learn invariant features.

- **Diffusion-Based Augmentation:** Synthetic images are generated using a class-conditional DDPM, trained either on the full dataset or on the subsets used to train the classifier. These images are generated separately for each class and incorporated into the training set in a manner consistent with the original class label distribution.

However, to gain a deeper understanding of how diffusion-based augmentation influences classifier performance, we explore the effects of different ratios of synthetic-to-real training data. Specifically, we consider scenarios where synthetic images are added in increasing amounts: **5%, 10%, 20%, 30%, 40%, 50%** of the original dataset size.

This allows us to assess whether there exists an optimal level of synthetic data, beyond which performance plateaus or degrades—potentially due to overfitting or domain shift. We present the results in section 5.

Following related literature, our key evaluation metric used is the classification accuracy.

5 Results

This section details the empirical outcomes of the experiments described in Section 4. We begin by presenting a comprehensive overview of the classification performance under various synthetic and geometric data augmentation strategies and across our varied training dataset sizes (100%, 50% and 10%). Our primary objective is to elucidate the effectiveness of synthetic data augmentation using a simple DDPM, and to ascertain whether different levels of augmentation can impact the performance. We use different dataset sizes because this application is particularly appropriate in cases of limited training data. We report our findings, and where possible provide robust statistical analyses.

(a) Baseline Performance and Comparative Effectiveness

Our experimental investigation quantified the impact of several augmentation strategies on classification accuracy for the EuroSAT dataset. Table 3 summarizes the test accuracies achieved across the three dataset conditions (100%, 50% and 10%) and the various augmentation approaches used.

We established initial performance benchmarks under the "No Augmentation" condition. With the complete (100%) dataset, a baseline test accuracy of 98.94% was achieved, indicating a high baseline model performance even with the unaltered dataset. Reducing the dataset size led to a corresponding decrease in baseline accuracy: 98.41% for the 50% dataset and 96.39% for the 10% dataset. We initially expected the 10% dataset to yield a lower baseline accuracy, as this would better reflect real-world scenarios with limited training data. However, the observed baseline performance was relatively high, reducing the potential impact of our augmentation strategies on improving accuracy. For the 50% and 10% dataset scenarios, where performance variability is typically higher, we conducted the experiments for multiple runs, and average test accuracies are reported (see Table 3) to obtain robust results. For the full (100%) dataset we could not perform these experiments due to computational limits and cost

constraints, hence we report only a single run, and the findings for this should be taken with a grain of salt.

Traditional geometric augmentation demonstrated a consistent ability to enhance performance across all dataset sizes:

- With the full dataset, geometric augmentation achieved the highest accuracy of 99.13%.
- For the 50% dataset geometric augmentation resulted in an average accuracy of 98.69%, representing a statistically significant improvement of 0.34 percentage points ($\Delta = +0.34$) compared to the baseline
- For the 10% dataset, geometric augmentation also had the most pronounced impact with an improvement of 0.79 percentage points ($\Delta = +0.79$, $p < 0.001$) over its 96.39% baseline.

These results highlight the general effectiveness of standard geometric transformations, particularly when the volume of training data is constrained.

The introduction of synthetic data yielded surprising, yet still slightly varied outcomes, with performance appearing sensitive to both the proportion of synthetic samples integrated and the size of the original dataset.

For the full (100% dataset, incorporating synthetic data at any tested percentage (5% to 50%)) did not improve upon the 98.94% no-augmentation baseline. Accuracies with synthetic data ranged from 98.61% (30% Synthetic) to 98.79% (5% Synthetic), generally registering marginally lower than the baseline.

When considering the 50% dataset:

- The addition of 5% synthetic data resulted in an accuracy of 98.22% ($\Delta = -0.12$ percentage points compared to the 98.41% baseline for this dataset size).
- Increasing proportions of synthetic data led to further reductions in mean accuracy relative to the no-augmentation baseline. For instance, 10% synthetic data yielded 98.06% ($\Delta = -0.29$ percentage points, $p < 0.001$), 20% synthetic data resulted in 98.02% ($\Delta = -0.32$ percentage points, $p < 0.001$), 30% synthetic data gave 97.84% ($\Delta = -0.51$ percentage points, $p < 0.001$), and 50% synthetic data achieved 97.81% ($\Delta = -0.53$ percentage points, $p < 0.001$). These decreases were statistically significant.

In the case of the 10% dataset, representing the most data-scarce scenario:

- Smaller percentages of synthetic data, such as 5% (96.24%, $\Delta = -0.15$ percentage points), 10% (96.27%, $\Delta = -0.12$ percentage points), and 20% (96.14%, $\Delta = -0.25$ percentage points), resulted in accuracies near the 96.41% no-augmentation baseline while these differences were not statistically significant.
- However, introducing larger proportions of synthetic data exhibited a more pronounced negative impact on accuracy. The inclusion of 30% synthetic data led to an accuracy of 94.86% ($\Delta = -1.53$ percentage points, $p < 0.001$), and 50% synthetic data resulted in 95.93% ($\Delta = -0.46$ percentage points, $p < 0.01$). Both

reductions were statistically significant compared to the baseline. The 40% synthetic data condition yielded 95.62% ($\Delta = -0.0.77$ percentage points). These findings suggest diminishing returns or even degradation when relying too heavily on synthetic samples.

However, it is important to note that the observed degradation in performance does not necessarily imply that synthetic augmentation is ineffective necessarily. Rather, it may reflect limitations in the quality or realism of the generated samples produced by our diffusion model. For instance, if the initial inclusion of 5% synthetic data had yielded clear improvements over the baseline, it could be that higher proportions would have amplified this positive effect. Instead, the synthetic data failed to outperform or meaningfully support the model even at low volumes, suggesting a potential mismatch between the synthetic and real data distributions. The reasons for this will be shortly analyzed in section (b)

Considering the significance levels, it is interesting to note that the results for the 10% dataset are more varied than those for the 50% case. This is expected, as the reduced amount of training data leads to higher variance in model performance across runs. As a result, statistical tests are less likely to detect significant differences, even when mean accuracies differ. Furthermore, training a model exclusively on 100% synthetic data (equivalent in size to the full dataset) resulted in an accuracy of only 83.59%, a substantial drop compared to training on real data, highlighting limitations in the synthetic data’s standalone quality or its distribution match with real data. This aligns with our earlier hypothesis that the quality of the synthetic samples, particularly when used in isolation, is insufficient to fully replicate the real-world data for our task. The pronounced drop in accuracy when training exclusively on synthetic samples further supports the notion that the generated data may suffer from distributional mismatches or overall lacking quality of the images. However, this does not imply that synthetic augmentation is universally ineffective. Its impact might be more nuanced (i.e offering benefits for specific land cover classes) despite not improving overall accuracy. We shall investigate this in more detail in the section (b).

The key observations from these experiments are — Geometric augmentation consistently achieved the highest test accuracies, with its benefits being statistically significant and more substantial as the size of the original dataset decreased. The use of synthetic data, within the parameters of our experiments, did not exceed the performance of the no-augmentation baseline. In many instances, particularly with larger percentages of synthetic data relative to the original data, or when the original dataset was smaller, its inclusion was associated with lower mean test accuracies, some of which were statistically significant.

Table 3: Test accuracy across augmentation strategies and dataset sizes

Strategy	100%	50%		10%	
	Acc.	Acc.	Δ	Acc.	Δ
No Augmentation	98.94	98.41	–	96.39	–
Geometric	99.13	98.69	+0.34***	97.18	+0.79***
5% Synthetic	98.79	98.22	–0.12	96.24	–0.15
10% Synthetic	98.75	98.06	–0.29***	96.27	–0.12
20% Synthetic	98.66	98.02	–0.32***	96.14	–0.25
30% Synthetic	98.61	97.84	–0.51***	94.86	–1.53***
40% Synthetic	98.64	97.99	–0.35***	95.93	–0.46 **
50% Synthetic	98.62	97.81	–0.53***	95.62	–0.77***
100% Only Synthetic	83.59	–	–	–	–

Note: Accuracy values for the 50% and 10% datasets are averages over 10 and 20 independent runs, respectively. Statistical comparisons are made against the no-augmentation baseline using Welch’s t-test.

Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Δ indicates the difference in mean accuracy relative to the no-augmentation baseline.

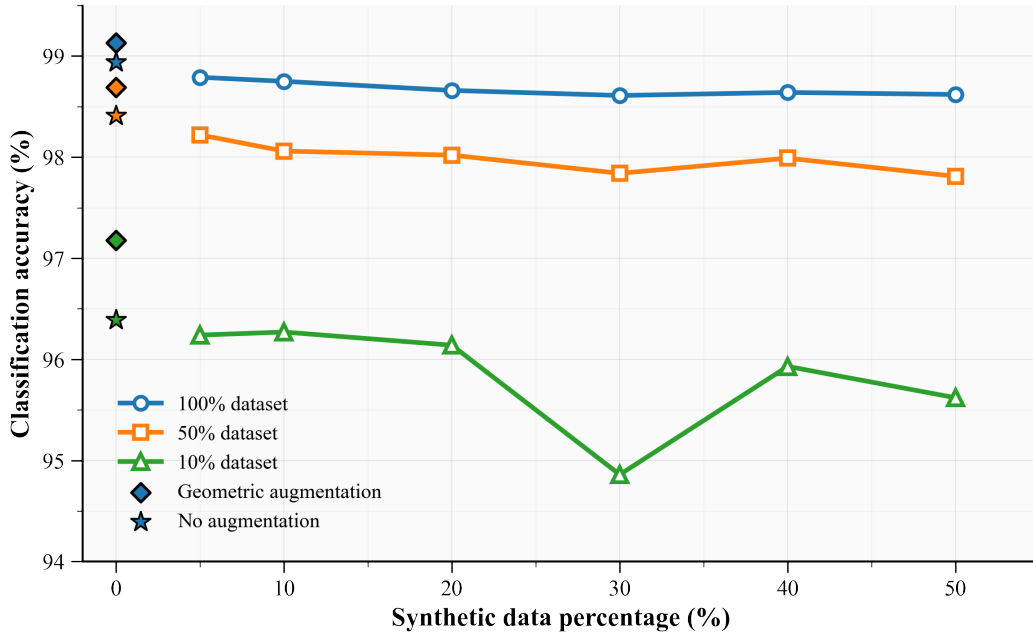


Figure 9: Comparison of test accuracy across different augmentation strategies. At $x=0$, the points show the baseline (no augmentation) performance, while the triangular markers represent geometric augmentation. The blue line represents results on the 100% dataset, while the orange line shows results on the 50% dataset.

As shown in Figure 9, geometric augmentation consistently outperformed other strategies, achieving the highest test accuracy for both dataset sizes.

(i) **Effectiveness of Hybrid Augmentation: Combining Synthetic and Geometric Approaches**

Following the evaluation of individual augmentation techniques, we investigated whether combining geometric transformations with DDPM-generated synthetic data could yield valuable benefits. Table 4 presents the test accuracy results for these hybrid approaches, compared against the no-augmentation and geometric-only baselines.

Table 4: Test accuracy and statistical significance of hybrid (Synthetic + Geometric) augmentation strategies

Strategy	100%	50%		10%	
	Acc.	Acc.	Δ	Acc.	Δ
No Augmentation	98.94	98.41	–	96.39	–
Geometric Augmentation	99.13	98.69	+0.34***	97.18	+0.79***
5% Synthetic + Geo Aug	98.91	98.66	+0.25**	96.78	+0.39**
10% Synthetic + Geo Aug	98.91	98.35	+0.06	96.77	+0.38**
20% Synthetic + Geo Aug	98.96	98.50	+0.09	96.56	+0.17
30% Synthetic + Geo Aug	98.96	98.35	-0.06	96.16	-0.23
40% Synthetic + Geo Aug	98.83	98.33	-0.06	96.71	+0.31*
50% Synthetic + Geo Aug	98.85	98.44	+0.03	96.67	+0.28

Note: 100% dataset results are single-run values. Accuracy values for the 50% and 10% datasets are averages over 10 and 20 independent runs, respectively. Statistical comparisons are made against the no-augmentation baseline using Welch’s t-test.

Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Δ indicates the difference in mean accuracy relative to the no-augmentation baseline.

The combined approach yielded several notable findings. For the reduced dataset (50% of original data) - in some runs - the addition of 5% synthetic data to geometric augmentation achieved the highest individual accuracy with improvements over the baseline and a marginal improvement over geometric augmentation alone. However, when averaged across multiple runs, this hybrid approach (98.66%) performed slightly below pure geometric augmentation (98.69%), indicating that while promising individual results can occur, the improvement is not consistently reproducible.

For the full dataset, synthetic data alone did not improve over the baseline. When combined with geometric augmentation, certain proportions yielded marginally better results than the baseline. Specifically, adding 20–30% synthetic data to geometric transformations achieved a test accuracy of 98.96%, which marginally exceeded the no-augmentation baseline (98.94%) but remained below pure geometric augmentation (99.13%). This is especially evident in the 10% dataset, where geometric augmentation achieved a significant gain of +0.79 percentage points ($p < 0.001$), outperforming all combined strategies including the best-performing hybrid setup (5% Synthetic + Geometric: +0.39 percentage points, $p < 0.01$). These results indicate that, in our setting, introducing synthetic data does not or only barely synergize with geometric augmentation.

These results highlight an important insight: while diffusion-generated images do not

consistently enhance model performance, especially when used alone, their selective integration with traditional augmentation techniques may offer marginal benefits under specific conditions.

(b) Per-Class Performance Analysis

To gain deeper insights into the effectiveness of different augmentation strategies, we conducted a detailed per-class analysis of the classification performance on the 10% dataset. This analysis reveals that while overall accuracy improvements may be modest, certain land cover classes benefit substantially from specific augmentation approaches, particularly hybrid combinations of synthetic and geometric augmentation.

(i) Class-Specific Optimal Strategies

Table 5 presents the optimal augmentation strategy for each land cover class based on F1-score performance. The results demonstrate strong class-specific preferences, with hybrid strategies dominating overall and a few classes preferring either geometric-only or synthetic-only augmentation.

Table 5: Best augmentation strategy by class with F1-score performance and improvement over baseline and 10 runs

Class	Best Strategy	F1 Score (%)	Improvement (pp)
AnnualCrop	40% Synthetic + Geometric	97.44	2.65
PermanentCrop	Geometric Augmentation	94.43	1.90
Pasture	Geometric Augmentation	96.84	1.37
Highway	10% Synthetic + Geometric	95.89	1.05
Industrial	5% Synthetic + Geometric	96.49	0.70
Residential	5% Synthetic Data	99.26	0.59
HerbaceousVegetation	20% Synthetic + Geometric	95.86	0.52
SeaLake	40% Synthetic Data	99.66	0.51
River	50% Synthetic + Geometric	98.07	0.36
Forest	Geometric Augmentation	98.98	0.26

Hybrid augmentation dominance: Five out of ten classes achieved their best performance using hybrid synthetic + geometric approaches:

- **AnnualCrop:** 40% Synthetic + Geometric (F1: 97.44%, +2.65 pp) – the largest single improvement observed
- **HerbaceousVegetation:** 20% Synthetic + Geometric (F1: 95.86%, +0.52 pp)
- **Highway:** 10% Synthetic + Geometric (F1: 95.89%, +1.05 pp)
- **Industrial:** 5% Synthetic + Geometric (F1: 96.49%, +0.70 pp)
- **River:** 50% Synthetic + Geometric (F1: 98.07%, +0.36 pp)

Pure synthetic data preference: Two classes performed optimally with synthetic data alone:

- **Residential:** 5% Synthetic Data (F1: 99.26%, +0.59 pp)
- **SeaLake:** 40% Synthetic Data (F1: 99.66%, +0.51 pp)

Geometric-only preference: Three classes performed optimally with pure geometric augmentation:

- **PermanentCrop:** Geometric Augmentation (F1: 94.43%, +1.90 pp)
- **Pasture:** Geometric Augmentation (F1: 96.84%, +1.37 pp)
- **Forest:** Geometric Augmentation (F1: 98.98%, +0.26 pp)

Variable synthetic ratios: The optimal synthetic data ratios varied significantly across classes, ranging from 5% to 50%, suggesting that different land cover types benefit from different levels of synthetic augmentation.

Performance gap analysis: An important pattern emerges when examining the performance differences between optimal and sub-optimal strategies. For classes where Synthetic + Geometric achieved the best performance (River, HerbaceousVegetation, Highway, Industrial, and AnnualCrop), pure geometric augmentation consistently ranked as the close second-best option, with only marginal performance gaps. This suggests that these classes benefit modestly but consistently from the addition of synthetic data to geometric transformations.

Conversely, for classes where geometric augmentation alone performed optimally (PermanentCrop, Pasture, and Forest), the hybrid approaches showed significantly lower performance, indicating that synthetic data introduction may actually hinder classification for these particular land cover types. This divergent behavior reinforces the class-dependent nature of synthetic augmentation effectiveness and suggests that certain land cover characteristics are better preserved through geometric transformations of real data rather than synthetic data generation.

Overall, these findings suggest that the effectiveness of augmentation is highly class-dependent. Especially hybrid strategies have the potential to lead to notable F1-score improvements in several classes.

(ii) Confusion Matrix Analysis

Figure 10 displays confusion matrices for four augmentation strategies applied to the 10% dataset: No Augmentation, Geometric Augmentation, 5% Synthetic Data, and 5% Synthetic + Geometric. All matrices exhibit strong diagonal dominance, with most classes achieving over 90% correct classification, underscoring a high baseline performance even with limited training data.

Importantly, averaging over 20 runs ensures that the observed differences are stable. While most classes maintain high prediction accuracy across all strategies, subtle differences become visible. Notably, **Geometric Augmentation** consistently yields the cleanest confusion matrix, with the most sharply concentrated diagonal and minimal off-diagonal noise. This reflects its strong generalization capability across both simple and challenging classes.

The Highway class, for example, shows an increase in correct predictions from 51 (no augmentation) to 52 (geometric augmentation / hybrid), with reduced confusion particularly with Forest. River classification also improves slightly (from 49 to 50), showing better separation from classes like AnnualCrop. Notably, the hybrid strategy minimizes these confusions more effectively than 5% synthetic alone, which occasionally increases misclassification (e.g., Herbaceous or Highway).

Furthermore, these improvements are often marginal—typically one or two instances per class—even when averaged over 20 runs. This reinforces earlier observations from F1-score metrics: while geometric augmentation offers consistent and interpretable gains, synthetic data—especially when used alone—rarely improves class separation and may sometimes increase confusion. For example, the 5% Synthetic condition shows slightly more off-diagonal dispersion in classes like *HerbaceousVegetation*, suggesting minor degrading of model performance or redundancy in the synthetic samples.

Industrial areas maintain similar performance across strategies (49–50 correct), but the geometric augmentation method again shows marginal gains by reducing overlap with Highway one of the most common errors due to their shared human-modified features. Likewise, PermanentCrop benefits from a small but stable increase in precision, with confusion from AnnualCrop and Herbaceous diminishing under geometric and hybrid strategies.

Across all strategies, confusion between visually and spectrally similar classes persists, such as Pasture and Herbaceous or Industrial and Residential. These persistent overlaps reflect intrinsic ambiguities in land cover distinctions, particularly in low-resolution or transition zones.

The hybrid 5% Synthetic + Geometric approach, while still performing well, does not surpass the pure geometric method in overall clarity or class-wise consistency. This reinforces that geometric transformations alone provide superior effectiveness compared to introducing synthetic images, which may add minor noise or create redundancy in class boundary decisions.

Overall, these averaged confusion matrices visually support the conclusion that **pure geometric augmentation remains the most reliable strategy**. Hybrid and synthetic-only approaches remain limited by the quality and utility of the generated samples but further refinement like class specific preferences could help in that aspect and might even be optimal.

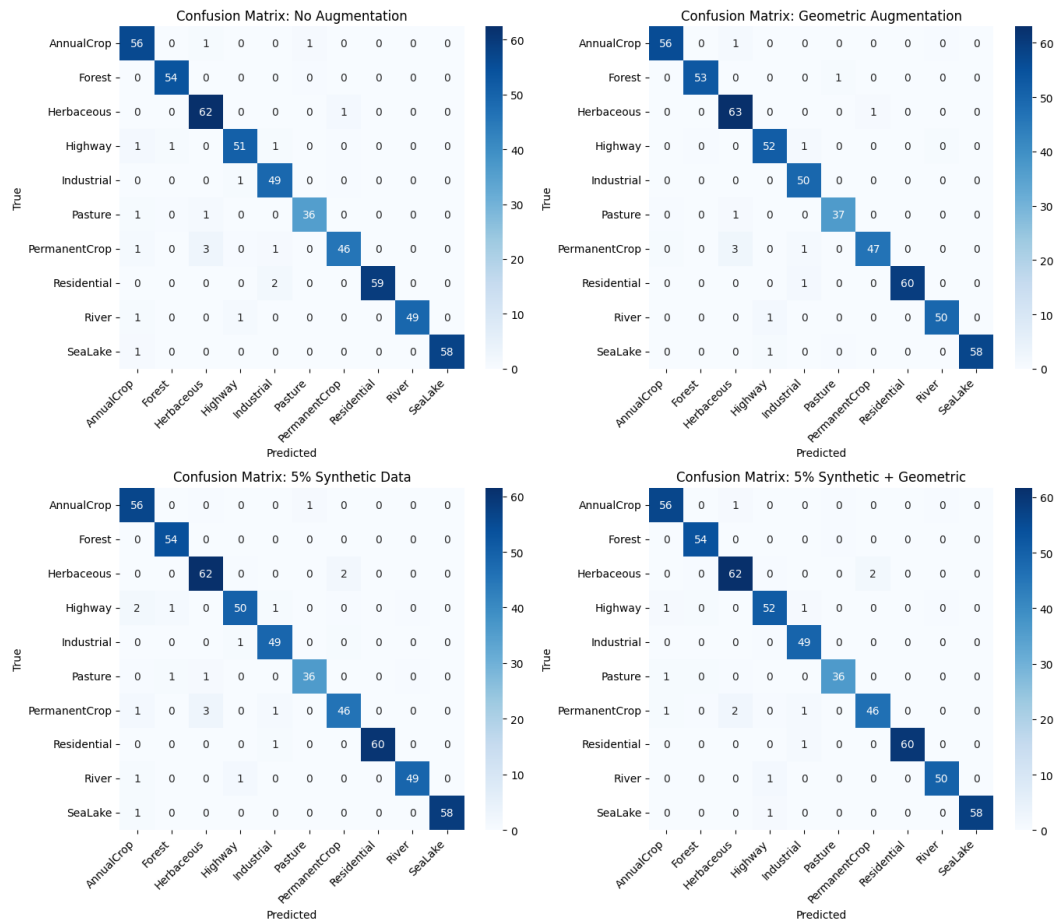


Figure 10: Confusion matrices with different augmentation methods, average over 10 runs

6 Discussion and Conclusion

This paper’s result highlights the potential but also the limitation of data augmentation. We have seen that geometric augmentation remains one of the most powerful tricks in this domain but that at the same time synthetic image creation can also help and their combined approach may even be the best method.

Nevertheless, geometric augmentation consistently outperformed other strategies across all data availability settings. Its simplicity, ease of implementation, and strong empirical performance make it highly reliable—especially when training data is limited. Averaged over 20 runs, geometric transformations yielded the most robust improvements in accuracy and F1-score, with cleaner confusion matrices and stronger class-wise separability.

On the other hand, diffusion-based synthetic augmentation led to stagnating or even deteriorating performance. This was particularly evident in the 10% and 50% dataset conditions, where larger synthetic ratios (30%–50%) resulted in statistically significant drops in accuracy.

This leads us to take a closer look at the potential reasons for these findings.

One possible explanation is that the generation of synthetic images may introduce noise or misleading patterns, particularly when class distinctions are subtle. The classifier may learn to rely on these artifacts, which exist only in the synthetic domain and are not truly representative of real data. As a result, generalization performance suffers, especially when synthetic images make up a large proportion of the training data, increasing the risk of overfitting to features that are artificial rather than meaningful.

Another plausible explanation relates to the data distribution learned by the diffusion model. Since the diffusion model is trained on the same dataset as the classifier (full or partial), it essentially captures and then reproduces the same underlying distribution. Consequently, the synthetic images do not introduce new information, but rather replicate what is already present in the original data. This limits the utility of synthetic augmentation, particularly in higher-data regimes, where the classifier already has a strong understanding of the distribution and gains little from additional, redundant samples. On top of that, the specific conditions/configuration of our DDPM may have played a role. Nowadays, synthetic data augmentation is traditionally done with text-to-image diffusion models, and hence the image quality may have been too capped for our DDPM compared to that of a SOTA diffusion model.

Furthermore, a well-performing classifier such as our classifier using 100% of the data may have already captured the key features necessary for the task. In this case, the model is close to its performance ceiling (99% accuracy), and additional training data; especially if it is generated from the same distribution—provides barely any additional information. This leads to something such as diminishing returns from data augmentation, as the marginal utility of each new sample decreases.

However, it is interesting to see that geometric augmentation consistently seems to add value to the model, no matter the dataset size and baseline accuracy. This might be due to the fact that geometric augmentation only works with real images, applying transformations such as flips on them. Importantly, the correct label is kept and no

noise in our augmentation pipeline is added. In the end, it helps learning more invariant and robust features without introducing new, potentially misleading patterns.

Synthetic image generation, on the other hand, does not work that way and there is the risk that the classifier learns wrong features that can potentially harm generalization as mentioned above.

Thus, geometric augmentation remains a bit more reliable than using DDPM as a generative model, but this does not mean that augmentation with synthetic images does not work: We saw how it also has the potential to improve accuracy. However, it should be interpreted with caution as it depends on multiple factors — including the specific train-test split, dataset size, class distribution, the fidelity of the synthetic images, and the training parameters of both the classifier and the diffusion model. Thus, while synthetic augmentation remains a promising technique—especially when paired with geometric transformations—its benefits are highly context-dependent. Future work may explore more targeted generation strategies (e.g., class-conditional synthesis, hard example mining, classifier-free-guidance) or integrate synthetic data selectively based on classifier uncertainty or feature diversity.

Ultimately, our findings suggest that geometric augmentation remains the most reliable and globally effective method in this setting. However, synthetic augmentation has clear potential, although its application must be carefully chosen to avoid counter-productive effects.

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